Holm's Step-Down Correction

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Introduction

Holm's (1979) step-down procedure controls the family-wise error rate (FWER) but is uniformly more powerful than Bonferroni. It tests the smallest p-value against α/m, and each subsequent p-value against a progressively larger threshold as the earlier hypotheses are rejected.

Prerequisites

Bonferroni correction, multiple testing.

Theory

Order the m p-values as $p_{(1)} \leq p_{(2)} \leq \ldots \leq p_{(m)}$. Define thresholds

$$\alpha_j = \alpha/(m - j + 1)$$

for $j = 1, \ldots, m$. Reject $H_{(1)}$ if $p_{(1)} \leq \alpha_1$; if so, reject $H_{(2)}$ if $p_{(2)} \leq \alpha_2$; continue sequentially. Stop at the first non-rejection; all subsequent hypotheses are also not rejected.

Equivalently, the Holm-adjusted p-value is

$$p_{(j)}^{\text{adj}} = \min\left[1, \max_{i \leq j}(m - i + 1)p_{(i)}\right].$$

Holm always rejects at least as many hypotheses as Bonferroni, strictly more when any rejection occurs. It controls FWER at α under any dependence structure.

Assumptions

Same as Bonferroni: no distributional assumptions on the p-values beyond validity.

R Implementation

```r
# Simulated p-values across 8 tests
p_raw <- c(0.001, 0.005, 0.012, 0.018, 0.025, 0.040, 0.080, 0.2)
m <- length(p_raw)

# Holm via p.adjust
p_holm <- p.adjust(p_raw, method = "holm")

# Bonferroni for comparison
p_bonf <- p.adjust(p_raw, method = "bonferroni")

data.frame(
  raw = p_raw,
```

```
  bonf = p_bonf,
  holm = p_holm,
  reject_at_0.05_holm = p_holm < 0.05,
  reject_at_0.05_bonf = p_bonf < 0.05
)
```

Output & Results

```
    raw  bonf   holm reject_at_0.05_holm reject_at_0.05_bonf
1 0.001 0.008 0.008                TRUE                TRUE
2 0.005 0.040 0.035                TRUE                TRUE
3 0.012 0.096 0.072               FALSE               FALSE
4 0.018 0.144 0.090               FALSE               FALSE
5 0.025 0.200 0.100               FALSE               FALSE
6 0.040 0.320 0.120               FALSE               FALSE
7 0.080 0.640 0.160               FALSE               FALSE
8 0.200 1.000 0.200               FALSE               FALSE
```

Both Holm and Bonferroni reject the same two hypotheses here. Holm's adjusted p-values are always $\leq$ Bonferroni's, opening space for additional rejections in other examples.

Interpretation

"After Holm step-down correction for multiple testing, two of the eight tests reached significance at family-wise $\alpha = 0.05$ (raw p = 0.001 and 0.005)."

Practical Tips

- Prefer Holm to Bonferroni by default; it is always at least as powerful and equally valid.
- Both procedures assume no prior ordering of hypotheses; if hypotheses are pre-ordered, fixed-sequence testing is more powerful.
- For very large m, FWER-controlling procedures are harsh; consider FDR methods (Benjamini-Hochberg) instead.
- Hochberg's step-up procedure is slightly more powerful than Holm but requires independence or positive dependence.
- Holm can be combined with post-hoc comparisons and with stratified tests without additional adjustment.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors

- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests